

Backpropagation in Deep Learning - Part 1 (The What)

1 What is Backpropagation?

Backpropagation (short for *backward propagation of errors*) is the **process by which a neural network learns** — it adjusts its internal parameters (weights and biases) to reduce the loss.

Think of it like a **teacher correcting a student's mistakes**:

- Forward pass → make a prediction
- Compare prediction to actual answer → find error (loss)
- Backward pass → send that error backward through the network to correct weights

That's backpropagation.

2 The Basic Idea

Neural networks learn using **two phases**:

(a) Forward Propagation

- Inputs go through layers → outputs → compute loss.
- This tells *how wrong* the network is.

(b) Backward Propagation

- The network looks at the loss and says:
“Okay, which weights caused this mistake?”
- Then it **updates those weights** slightly to reduce the error next time.

This “sending the blame backward” is why it's called *back-propagation*.

3 Why We Need It

In a large neural network, there are **thousands or millions of weights**.

We can't manually adjust them — so backpropagation uses **calculus (derivatives)** to compute exactly **how much each weight contributed to the error**.

That derivative is the **gradient** (slope).

4 Step-by-Step Intuition

Let's imagine a small example:

Step 1: Forward Pass

- You feed in inputs → the model gives predictions.
- Compute loss = (actual – predicted) difference.

Step 2: Compute Error

- Calculate how much each output neuron's prediction was off.

Step 3: Send Error Back

- Send this “error signal” backward through the network, layer by layer.
- Each neuron learns how much it contributed to the total error.

Step 4: Update Weights

- Adjust each weight slightly:
$$\text{new weight} = \text{old weight} - \text{learning rate} \times \text{gradient}$$
- This makes the next prediction a bit better.

5 Real-World Analogy

Imagine you’re learning to throw darts:

- First throw → you miss → measure *how far* and *in what direction* you missed (loss).
- You adjust your hand slightly → throw again.
- Repeat until your throws are accurate.

That’s what backpropagation does — small adjustments over many rounds (epochs) until the model “hits the target”.

6 Key Components

Concept	Meaning
Loss Function	Measures how wrong the model is
Gradient	Direction + magnitude of change needed
Learning Rate	How big each update step should be
Weight Update Rule	$w = w - \eta \cdot \frac{\partial L}{\partial w}$
Chain Rule	Mathematical trick to send error backward through layers

7 What Makes It Powerful

- It works for **any number of layers** (deep networks).
- It automatically distributes the error signal across layers.
- It’s used with **gradient descent** for optimization.
- It’s the foundation of **all modern deep learning models** (CNNs, RNNs, Transformers, etc.).

In Summary

Step	Process	Purpose
1	Forward Pass	Make predictions
2	Compute Loss	Measure error

3	Backward Pass	Compute gradients
4	Weight Update	Learn and improve

→ **Backpropagation = How neural networks learn from mistakes.**
